

sanity check: Top regions by agricultural percent

```
import numpy as np
import pandas as pd
import geopandas as gpd
import os.path as path
import utils
import matplotlib.pyplot as plt
from collections import Counter
```

```
lau_df = utils.load_complex_geojson(path.join(utils.out_data_dir, "drought_days_lau.geojson"))
nuts3_df = utils.load_complex_geojson(path.join(utils.out_data_dir, "nuts3_stats.geojson"))
country_df = pd.read_json(path.join(utils.out_data_dir, "country_stats.geojson"))
```

```
# all percentages are between 0 and 100
assert((nuts3_df["cropland_area_percent"].dropna() <= 100).all())
assert((nuts3_df["cropland_area_percent"].dropna() >= 0).all())
```

```
# all computed drought day values must be between 0 and 365
for col in ["max_drought_days", "median_drought_days"]:
    for temp_df in [nuts3_df, lau_df]:
        assert (temp_df[col].dropna() < 365).all()
        assert (temp_df[col].dropna() >= 0).all()
```

```
actual_countries = list(set(nuts3_df["cntr_code"].values))
# All EU + EEA country codes and the UK. From here:
# https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Country_codes
eea_countries = [
    "BE",
    "BG",
    "CZ",
```

```

    "DK",
    "DE",
    "EE",
    "IE",
    "EL",
    "ES",
    "FR",
    "HR",
    "IT",
    "CY",
    "LV",
    "LT",
    "LU",
    "HU",
    "MT",
    "NL",
    "AT",
    "PL",
    "PT",
    "RO",
    "SI",
    "SK",
    "FI",
    "SE",
    "IS",
    "LI",
    "NO",
    "CH",
]
assert len(eea_countries) == 30

other_countries = [
    "UK", # uk, manually added
    "RS", # slovakia, albania, north macedonia are in the NUTS and LAU dataset, despite not b
    "AL",
    "MK",
]
expected_countries = eea_countries + other_countries

assert expected_countries == other_countries

```

AssertionError:

AssertionError

Traceback (most recent call last)

Cell In[5], line 37

```
2 # All EU + EEA country codes and the UK. From here:
3 # https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Country
4 eea_countries = [
5     "BE",
6     "BG",
7     (...),
8     "CH",
9 ]
--> 37 assert len(eea_countries) == 30
39 other_countries = [
40     "UK", # uk, manually added
41     "RS", # slovakia, albania, north macedonia are in the NUTS and LAU dataset, desp
42     "AL",
43     "MK",
44 ]
45 expected_countries = eea_countries + other_countries
```

AssertionError:

nuts3_df.dropna(subset="cropland_area_percent").sort_values(by="cropland_area_percent", asce

	nuts_id	cntr_code	name_latn	nuts_name	area_km2	population	cropland_
1186	RS125	RS	Severnobačka oblast		1784.32	157170	1414.00
1164	RO312	RO	Călărași	Călărași	5078.66	278602 e	4011.83
1167	RO315	RO	Ialomița	Ialomița	4455.58	248786 e	3498.19
1169	RO317	RO	Teleorman	Teleorman	5786.99	313033 e	4358.58
549	DEG0D	DE	Sömmerda	Sömmerda	805.78	67693	589.23
...	...	...	...	...	...	...	...
853	IS002	IS	Landsbyggð	Landsbyggð	101650.66	140390	15.19
1033	NO074	NO	Troms og Finnmark	Troms og Finnmark	74774.22	242452	6.37
40	AT331	AT	Außerfern	Außerfern	1237.31	33946	0.00
703	FI1D7	FI	Lappi	Lappi	98963.93	176151	1.63
852	IS001	IS	Höfuðborgarsvæði	Höfuðborgarsvæði	1046.05	249054	0.01

Severnobačka oblast - The north of Serbia is dominated by agriculture, as seen on [satellite](#)

Călărași - is almost exclusively fields as confirmed visually on the areas around the city on [google maps](#)

More Romanian municipalities follow, which all reflect a highly agriculturally intensive country. See below

Sömmerda in Germany makes the list with over 70% agricultural area, which also checks out when looking at [satellite data](#)

Bottom regions by agricultural percent

Höfuðborgarsvæði - capital region of Iceland containing Reykjavik, seems plausible

Lappi - Northernmost parts of Finland are hostile to many crops

Außerfern - high up in the [austrian alps](#)

Troms og Finnmark - also known as Nordkap in the far north of Norway, checks out.

General country-level cropland stats

Note: we expect our data to underestimate all cropland statistics. The High resolution layer croplands does not account for fallow lands (Brachland), which varies strongly between years. Before 2024, at least 4% of all agricultural area were required to lie fallow [as per EU rules](#).

```
# https://www.destatis.de/DE/Themen/Branchen-Unternehmen/Landwirtschaft-Forstwirtschaft-Fischerei/Landwirtschaft-Forstwirtschaft-Fischerei/_inhalt.html
# UN stat: Cropland Area in the EU is 118810 km^2
nuts3 df[nuts3 df["cntr_code"] == "DE"]["cropland_km2"].sum()
```

```
np.float64(112229.78)
```

```
# https://statbase.org/data/rou-cropland/
# UN stat: 88130 # TODO: this discrepancy seems too high
nuts3_df[nuts3_df["cntr_code"] == "RO"]["cropland_km2"].sum()
```

```
np.float64(76041.45999999999)
```

```
# https://statbase.org/data/bgr-cropland/
# UN stat: 36225
nuts3_df[nuts3_df["cntr_code"] == "BG"]["cropland_km2"].sum()
```

```
np.float64(34389.3)
```

Sanity checking hotspots on the map

- Central Poland had a severe dry spell in 2019 about which a [case study](#) was published
- The border between CH and FR along the river Doubs is also [frequently struck](#)
- Almeria and Granada in southern Spain are known [drought hotspots](#)
- The 2025 spring drought in Sweden is also [reported on](#)

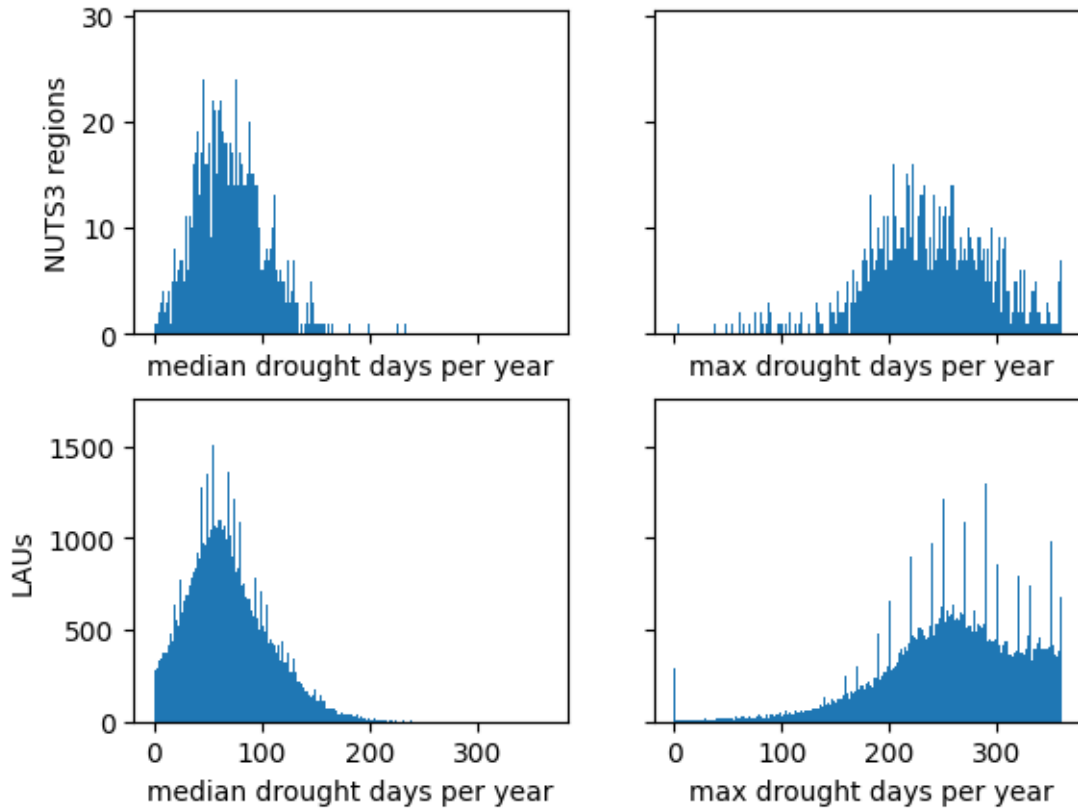
Sanity check: distribution of median and max drought days in LAUs and NUTS-3

```
fig, axs = plt.subplots(2, 2, sharex=True, sharey="row")

for i, metric in enumerate(["median", "max"]):
    axs[0][i].hist(nuts3_df[f"{metric}_drought_days"].values, bins=365, range=(0, 366) )
    axs[0][i].set_xlabel(f"{metric} drought days per year")

    axs[1][i].hist(lau_df[f"{metric}_drought_days"].values, bins=365, range=(0, 366) )
    axs[1][i].set_xlabel(f"{metric} drought days per year")

axs[0][0].set_ylabel("NUTS3 regions")
axs[1][0].set_ylabel("LAUs")
plt.show()
```



```
plt.hist(nuts3_df["cropland_area_percent"].values, bins=200)
plt.show()
```

```
(array([58., 23., 20., 23., 21., 15., 20., 16., 22., 16., 11., 15., 14.,
       12., 16., 22., 17., 16., 11., 21.,  8., 16., 19., 16., 11., 18.,
       8., 15., 15.,  6., 14., 13., 14., 11., 20., 10., 17., 10.,  9.,
       14., 20., 13.,  8., 17., 17.,  6., 14., 12., 11., 14., 14., 11.,
       11., 11.,  6., 11.,  9., 11., 12., 10.,  7., 12., 11., 12., 10.,
       10., 13.,  3., 13.,  8.,  7.,  6.,  5., 11.,  8., 11.,  6., 12.,
       6.,  5., 10., 13., 13.,  7.,  8.,  6.,  4.,  9.,  5., 10.,  7.,
       8.,  5.,  5.,  2.,  7.,  7.,  6.,  5.,  8.,  9.,  4.,  3.,  2.,
       5.,  2.,  5.,  5.,  0.,  4.,  6.,  3.,  3.,  3.,  4.,  5.,  2.,
       3.,  1.,  3.,  2.,  1.,  2.,  4.,  1.,  0.,  2.,  1.,  0.,  1.,
       1.,  5.,  2.,  1.,  0.,  0.,  0.,  2.,  1.,  2.,  1.,  2.,  1.,
       3.,  2.,  0.,  2.,  1.,  0.,  0.,  0.,  1.,  0.,  0.,  0.,  0.,
       0.,  1.,  1.,  1.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,
       0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,
       0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.,  0.]
```

```

0., 1., 1., 0., 2.]),
array([ 0.      ,  0.4974,  0.9948,  1.4922,  1.9896,  2.487 ,  2.9844,
        3.4818,  3.9792,  4.4766,  4.974 ,  5.4714,  5.9688,  6.4662,
        6.9636,  7.461 ,  7.9584,  8.4558,  8.9532,  9.4506,  9.948 ,
       10.4454, 10.9428, 11.4402, 11.9376, 12.435 , 12.9324, 13.4298,
       13.9272, 14.4246, 14.922 , 15.4194, 15.9168, 16.4142, 16.9116,
       17.409 , 17.9064, 18.4038, 18.9012, 19.3986, 19.896 , 20.3934,
       20.8908, 21.3882, 21.8856, 22.383 , 22.8804, 23.3778, 23.8752,
       24.3726, 24.87 , 25.3674, 25.8648, 26.3622, 26.8596, 27.357 ,
       27.8544, 28.3518, 28.8492, 29.3466, 29.844 , 30.3414, 30.8388,
       31.3362, 31.8336, 32.331 , 32.8284, 33.3258, 33.8232, 34.3206,
       34.818 , 35.3154, 35.8128, 36.3102, 36.8076, 37.305 , 37.8024,
       38.2998, 38.7972, 39.2946, 39.792 , 40.2894, 40.7868, 41.2842,
       41.7816, 42.279 , 42.7764, 43.2738, 43.7712, 44.2686, 44.766 ,
       45.2634, 45.7608, 46.2582, 46.7556, 47.253 , 47.7504, 48.2478,
       48.7452, 49.2426, 49.74 , 50.2374, 50.7348, 51.2322, 51.7296,
       52.227 , 52.7244, 53.2218, 53.7192, 54.2166, 54.714 , 55.2114,
       55.7088, 56.2062, 56.7036, 57.201 , 57.6984, 58.1958, 58.6932,
       59.1906, 59.688 , 60.1854, 60.6828, 61.1802, 61.6776, 62.175 ,
       62.6724, 63.1698, 63.6672, 64.1646, 64.662 , 65.1594, 65.6568,
       66.1542, 66.6516, 67.149 , 67.6464, 68.1438, 68.6412, 69.1386,
       69.636 , 70.1334, 70.6308, 71.1282, 71.6256, 72.123 , 72.6204,
       73.1178, 73.6152, 74.1126, 74.61 , 75.1074, 75.6048, 76.1022,
       76.5996, 77.097 , 77.5944, 78.0918, 78.5892, 79.0866, 79.584 ,
       80.0814, 80.5788, 81.0762, 81.5736, 82.071 , 82.5684, 83.0658,
       83.5632, 84.0606, 84.558 , 85.0554, 85.5528, 86.0502, 86.5476,
       87.045 , 87.5424, 88.0398, 88.5372, 89.0346, 89.532 , 90.0294,
       90.5268, 91.0242, 91.5216, 92.019 , 92.5164, 93.0138, 93.5112,
       94.0086, 94.506 , 95.0034, 95.5008, 95.9982, 96.4956, 96.993 ,
       97.4904, 97.9878, 98.4852, 98.9826, 99.48  ]),
<BarContainer object of 200 artists>)

```

